

18.600: Lecture 6

Conditional probability

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Outline

Definition: probability of A given B

Examples

Multiplication rule

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- ▶ Definition makes sense even without "equally likely" assumption.

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- ▶ Probability plane will come eventually, given plane not here yet.

Another famous Tversky/Kahneman study (Wikipedia)

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- ▶ Study participants believe blue taxi at fault, say witness correct with 80 percent probability.

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$$\begin{aligned} \blacktriangleright P(E_1 E_2 E_3 \dots E_n) = \\ P(E_1)P(E_2|E_1)P(E_3|E_1 E_2) \dots P(E_n|E_1 \dots E_{n-1}) \end{aligned}$$

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- ▶ Another example: roll die and let E_i be event that the roll *does not lie* in $\{1, 2, \dots, i\}$. Then $P(E_i) = (6 - i)/6$ for $i \in \{1, 2, \dots, 6\}$.

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- ▶ What is $P(E_4|E_1 E_2 E_3)$ in this case?

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- ▶ We have $P((1, 2)) = P((1, 3)) = 1/6$ and $P((2, 3)) = P((3, 2)) = 1/3$. Given host points to door 2, probability prize behind 3 is $2/3$.

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- ▶ Easy to see answer is $13/27$.